



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Advanced)
REVIEW TEST
26-06-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : ALL STAR BATCHES)

PAPER-1

Time : 2:45 Hours

Maximum Marks : 240

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **32** pages and that all the **20** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

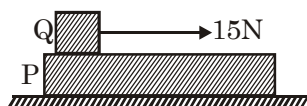
BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I : (Maximum Marks: 40)

- This section contains **TEN** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

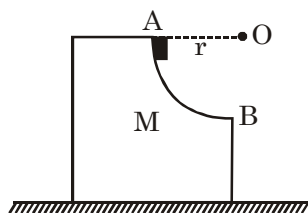
1. A long plank P of the mass 5 kg is placed on a smooth floor. On P, is placed a block Q of mass 2 kg. The coefficient of friction between P and Q is 0.5. If a horizontal force of 15N is applied to Q, then



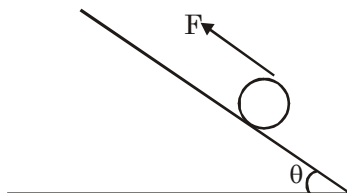
- (A) The total contact force on Q due to P is 10N.
- (B) The acceleration of Q relative to P is 2.5 m/s^2 .
- (C) The acceleration of P relative to the floor is 2.0 m/s^2 .
- (D) The acceleration of centre of mass of (P + Q) system relative to the floor is $(15/7) \text{ m/s}^2$.

Space for Rough Work

2. A small mass m slides from a point A of smooth quarter circular surface of mass M (free to move) as shown in figure. Mass M lies on smooth ground. When m reaches point B of M , then (AB is one fourth part of a circle of radius ' r '). [Line OA is horizontal, and OB is vertical]



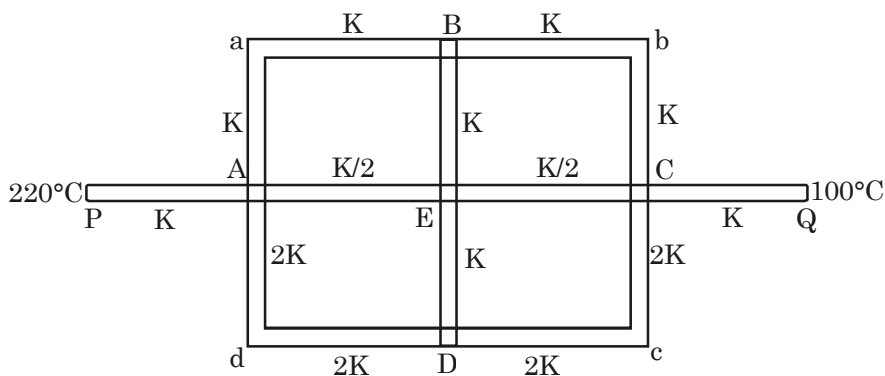
- (A) velocity of m w.r.t. M is $\sqrt{2gr}$.
 (B) velocity of M w.r.t. ground is $\frac{m\sqrt{2gr}}{M+m}$.
 (C) work done by normal reaction on m is zero.
 (D) none of these
3. A disc of mass M and radius R is placed on a rough incline of angle θ . A force is to be applied parallel to incline at the top most point of disc to keep it at rest.



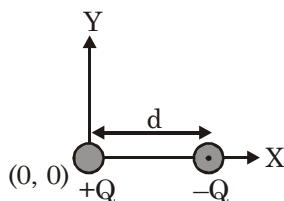
- (A) Applied force is $\frac{Mg \sin \theta}{2}$.
 (B) Friction force is $\frac{Mg \sin \theta}{2}$.
 (C) Minimum friction coefficient required to keep the body at rest in this situation is $\frac{\tan \theta}{2}$.
 (D) Minimum friction coefficient required to keep the body at rest in this situation is $\tan \theta$.

Space for Rough Work

4. The figure shows an arrangement of fourteen rods of same length ' ℓ ' and cross-sectional area ' A ' but with different conductivities as indicated. The ends P and Q are maintained at constant temperature of 220°C and 100°C under steady state condition. Find the temperature of the junctions A, B, C, D and E.



- (A) $T_A = 180$ (B) $T_B = 160$ (C) $T_C = 140$ (D) $T_E = 160$
5. Shown below is an electric dipole with equal charges $+Q$ and $-Q$ separated by a distance d . The dipole is free to rotate or move. Consider the following information:



- [I] The dipole does not feel a net torque about the axis of the dipole.
 [II] The dipole feels a net force in the $+x$ -direction.
 [III] When rotated from its original orientation and released, the dipole moves back towards the original orientation.

Which of the following electric field, can justify the above information, if $E_0 > 0$.

- (A) $\frac{E_0 x}{d}(-\hat{i})$ (B) $\frac{E_0 \left(x - \frac{3d}{4}\right)}{d}(-\hat{i})$ (C) $\frac{E_0 \left(x - \frac{d}{4}\right)}{d}(-\hat{i})$ (D) $\frac{E_0 x}{d}(\hat{i})$

Space for Rough Work

6. A proton is moving in gravity free space having a uniform electric field, E , of unknown magnitude and direction. There is a uniform magnetic field of 0.01 T pointing in the +y direction. The proton moves in the +x direction at a constant velocity of 10 km/s. All directions are with respect to the usual right-handed x-y-z coordinate system. Which of the following is/are **CORRECT** ?
- (A) $E_z = 100 \text{ V/m } \hat{k}$ (B) $E_z = -100 \text{ V/m } \hat{k}$
(C) $E_x = 100 \text{ V/m } \hat{i}$ (D) $E_x = 0 \hat{i}$
7. An α particle with a kinetic energy of 2.1 eV makes a head on collision with a hydrogen atom moving towards it with a kinetic energy of 8.4 eV. As a result,
- (A) Collision may be elastic
(B) Collision may be inelastic and a photon of energy 10.2 eV may be emitted
(C) Collision will be elastic
(D) Collision may be inelastic and a photon of energy 0.3 eV may be emitted

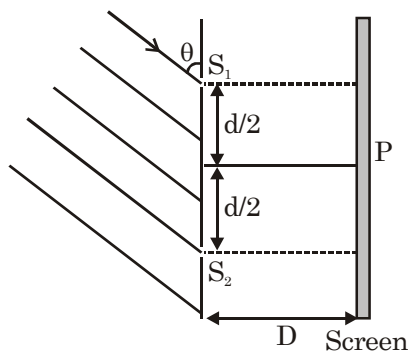
Space for Rough Work

8. A diverging lens of focal length f_1 is placed in front of and coaxially with a concave mirror of focal length f_2 . Their separation is d . A parallel beam of light incident on the lens returns as a parallel beam from the arrangement.
- (A) The beam diameters of the incident and reflected beams will not be same
(B) $d = 2|f_2| - |f_1|$
(C) If the entire arrangement is immersed in water, the value of d will remain unaltered.
(D) none of these
9. In measuring time period of simple pendulum, the maximum possible error in time period is reported as
- (A) the leastcount of stopwatch
(B) the average reaction time
(C) the minimum among leastcount of stopwatch and the reaction time
(D) the maximum among leastcount of stopwatch and the reaction time

Space for Rough Work

ALLEN

10. In a Young's double slit experiment, both the slits are to be covered with glass plates of equal thickness but with refractive index 1.5 and 1.2 respectively. The central maxima is desired to be observed at point P (as shown in the figure). The thickness of each plate is t .



- (A) Upper slit should have glass plate of refractive index 1.5.
 (B) Upper slit should have glass plate of refractive index 1.2.
 (C) $\theta = \cos^{-1} \left(\frac{3t}{10d} \right)$
 (D) $\theta = \sin^{-1} \left(\frac{3t}{10d} \right)$

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

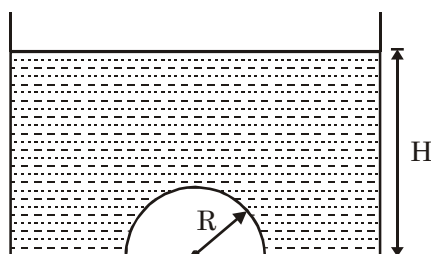
+									
●	●	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	●	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	●	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	●	●	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

-									
●	●	●	0	0	●	●	●	●	●
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	●	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	●	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

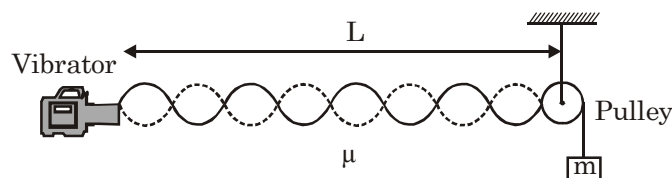
1. A hole in the horizontal bottom of the container is closed by a hemispherical cap of radius R (see figure). The vessel is filled with liquid of density d . Bottom is at a depth of H . Find the force with which the cap presses down on the bottom of the vessel. Neglect atmospheric pressure. If the force is ' $a\pi$ ' N, fill 'a' in the OMR sheet.

(Take : $d = 10^3 \text{ kg/m}^3$, $H = 1\text{m}$, $R = 30 \text{ cm}$)

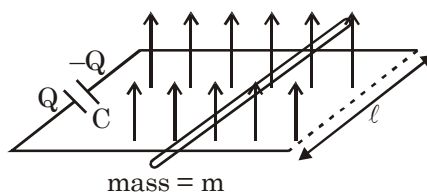


Space for Rough Work

2. In the arrangement shown in figure, a mass can be hung from a string (with linear mass density of $\mu = 0.002 \text{ kg/m}$) that passes over a light pulley. The string is connected to a vibrator (of constant frequency f), and the length of the string between point P and the pulley is $L = 2.00 \text{ m}$. When the mass m is either 9.0 kg or 25.0 kg , standing waves are observed. ($g = 9.8 \text{ m/s}^2$). What is the frequency (in Hz) of the vibrator?

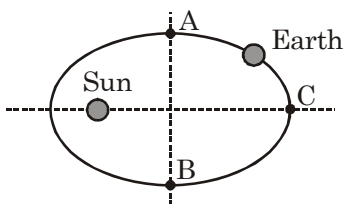


3. An arrangement of smooth pair of long conducting rails which are resistanceless are joined to a charged capacitor. The rails are connected by a rod of mass m and length ℓ and resistance R . Vertical magnetic field is uniform. The rod is initially at rest. It starts moving due to the influence of magnetic field. What should be the mass of the rod (in mg) so that its kinetic energy in steady state is maximum? [Take : $\ell = 1 \text{ m}$, $B = 2 \text{ T}$, $C = 1 \mu\text{F}$, $R = 1 \Omega$]

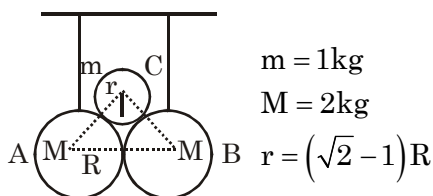


Space for Rough Work

4. Earth revolves around Sun with time period of ' $T = 365$ days'. In the figure shown below, find the time taken (in days) by the Earth to move from A to B in the path ACB if the eccentricity of ellipse is 0.5.

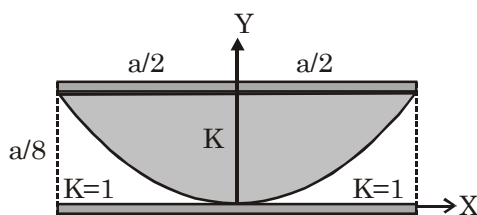


5. Two identical balls A & B each of mass 2 kg & radius R are suspended vertically from inextensible strings as shown. Third ball C of mass 1 kg & radius $r = (\sqrt{2} - 1)R$ falls & hits A & B symmetrically with 10 m/s . Speed of both A & B just after the collision is 3 m/s . Find speed of C (in m/s) just after collision.

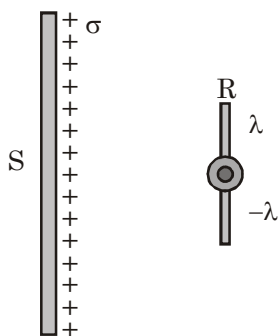


Space for Rough Work

6. A parallel plate capacitor consists of two square plates, each edge 'a'. There is a dielectric slab of dielectric constant K in the shape of a cylinder of parabolic cross section between the plates. The figure shows the cross section parallel to an edge of the plate. Find the capacitance of the capacitor (in fF). (Take $K = 2$, $a = 1$ mm).

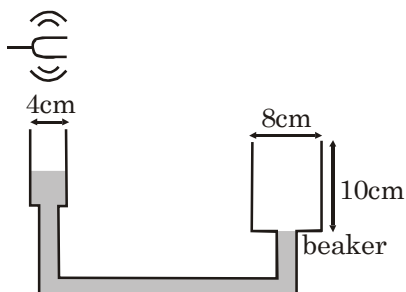


7. In the figure shown S is a large non-conducting sheet of uniform charge density σ . A rod R of length ℓ and mass 'm' is parallel to the sheet and hinged at its mid-point. The linear charge densities on the upper and lower half of the rod are shown in the figure. Find the angular acceleration (in radian/s²) of the rod just after it is released. (Take : $\lambda = \frac{1}{\pi} \mu\text{C}/\text{m}$, $\sigma = 1\text{mC}/\text{m}^2$, $m = 2\text{kg}$)



Space for Rough Work

8. In a resonance column apparatus, first resonance is obtained when the water filling beaker (of cylindrical shape) is just empty as shown. The water filling beaker is lowered down and it is seen that second resonance is obtained when beaker is filled upto brim. If the tuning fork has a frequency 420 Hz, what is the speed (in m/s) of sound in air ?



Space for Rough Work

SECTION-III : Integer Value Correct Type
No question will be asked in section III

SECTION-IV : (Maximum Marks : 16)

- This section contains **TWO** questions.
- Each question contains two columns, **Column-I** and **Column-II**.
- **Column-I** has **four** entries (A), (B), (C) and (D)
- **Column-II** has **five** entries (P), (Q), (R), (S) and (T)
- Match the entries in **Column-I** with the entries in **column-II**.
- One or more entries in **Column-I** may match with one or more entries in **Column-II**.
- The ORS contains a 4×5 matrix whose layout will be similar to the one shown below :

(A) ☐ (P) ☐ (Q) ☐ (R) ☐ (S) ☐ (T)

(B) ☐ (P) ☐ (Q) ☐ (R) ☐ (S) ☐ (T)

(C) ☐ (P) ☐ (Q) ☐ (R) ☐ (S) ☐ (T)

(D) ☐ (P) ☐ (Q) ☐ (R) ☐ (S) ☐ (T)

- For each entry in **column-I**, darken the bubbles of all the matching entries. For example, if entry (A) in **Column-I** matches with entries (Q), (R) and (T), then darken these three bubbles in the ORS. Similarly, for entries (B), (C) and (D).
- For each question, marks will be awarded in one of the following categories :

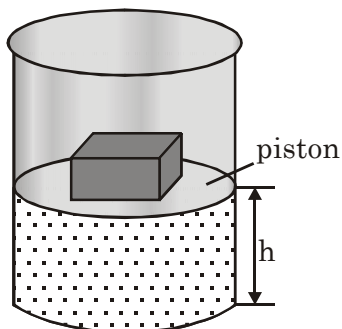
For each entry in **Column-I**

Full Marks : +2 If only the bubble(s) corresponding to all the correct match(es) is (are) darkened

Zero Marks : 0 If none of the bubbles is darkened

Negative Marks : -1 In all other cases

1. Hydrogen gas is kept in a container fitted with a piston of mass m . Many processes can be carried out on the gas. Match the possible outcome of each of the processes on the gas in the container.



Column-I

- (A) Some gas is slowly removed from the container keeping piston free. (container is highly conducting)
- (B) The gas is heated keeping the piston fixed (container is insulating)
- (C) The container is heated keeping the piston free (container is highly conducting)
- (D) The container is taken to a place where acceleration due to gravity is half of that on the earth but the temperature and pressure of the surroundings is the same. Piston is free. (container is highly conducting)

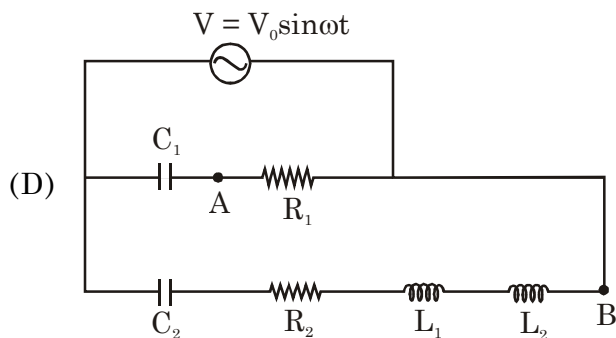
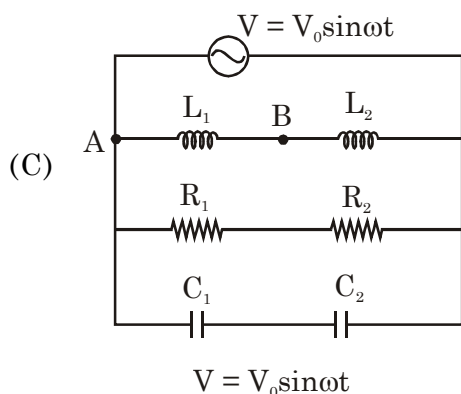
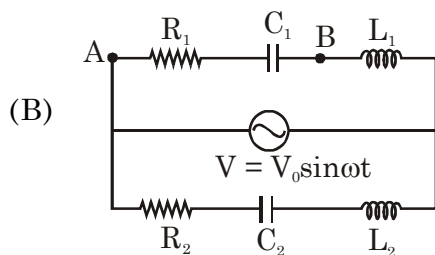
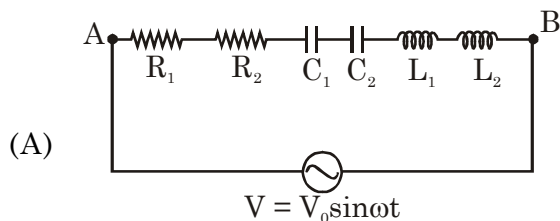
Column-II

- (P) The pressure of the gas increases
- (Q) The volume of the gas increases
- (R) The pressure of the gas decreases
- (S) The volume of the gas decreases
- (T) The temperature of the gas is the same as before.

Space for Rough Work

ALLEN

2. Circuits shown in column-I have two resistors R_1 and R_2 , two capacitors of capacitance C_1 and C_2 , and two inductors of self-inductance L_1 and L_2 . The circuits are connected to an AC source of voltage $V = V_0 \sin \omega t$ where t is the time. $V_{L_1}, V_{L_2}, V_{C_1}, V_{C_2}, V_{R_1}, V_{R_2}$ are instantaneous voltages across inductors, capacitors and resistors respectively.

Column-I**Column-II**

(P) V_{L_1} must be 180° ahead of V_{C_1} .

(Q) Instantaneous potential difference between A and B (V_{AB}) must be in phase with applied voltage.

(R) If $R_1 = R_2$, V_{R_1} is equal to V_{R_2} (in magnitude)

(S) V_{L_1} must be 180° ahead of V_{C_2} .

(T) V_{L_1} & V_{C_2} must be in phase

Space for Rough Work

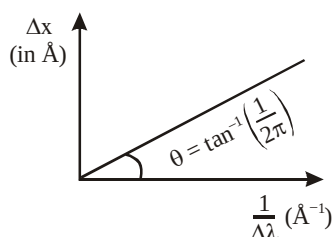
PART-2 : CHEMISTRY**SECTION-I : (Maximum Marks: 40)**

- This section contains **TEN** questions.
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1. A container contains equal mass of Helium and methane gases. A small pinhole is made in the container. Which of the following is/are not the ratio of mass of Helium to mass of methane effusing out initially ?

(A) $\frac{8}{1}$ (B) $\frac{1}{8}$ (C) $\frac{2}{1}$ (D) $\frac{1}{2}$

2. An electron is accelerated from rest, by 'V' volts. Uncertainty in position (Δx) is plotted against minimum uncertainty in de-broglie wavelength ($\Delta \lambda$) as following -



The value of 'V' is -

(A) 75 (B) 37.5 (C) 150 (D) 7.5

Space for Rough Work

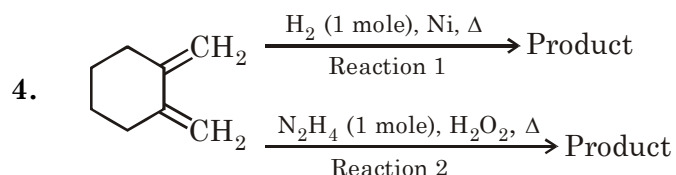
3. In FCC crystal, identify the correct statement(s) {a = edge length of unit cell }

(A) Size of octahedral void is more than that of tetrahedral void

(B) Shortest distance between two tetrahedral voids is $\frac{\sqrt{3}a}{2}$

(C) Shortest distance between two octahedral voids is $\frac{\sqrt{2}a}{2}$

(D) Tetrahedral void is surrounded by 6 sphere.



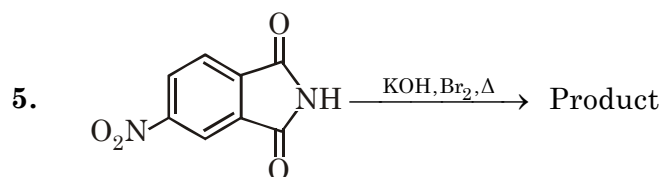
Correct statement(s) about major product(s) of reaction (1) and reaction (2) is/are :

(A) Product of reaction (1) is optically inactive

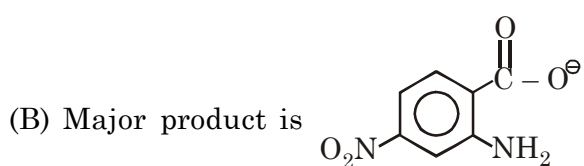
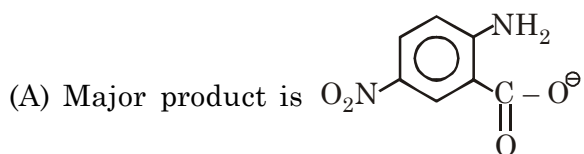
(B) Product of reaction (2) is enantiomeric pair

(C) Reaction (1) is 1,2-Addition

(D) Reaction (2) is 1,4-Addition



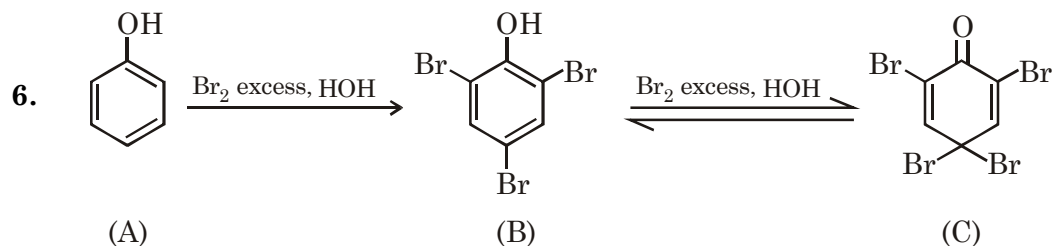
Correct statement(s) about reaction or product is/are :



(C) Reaction involve intramolecular rearrangement

(D) It involve acid base and $\text{S}_{\text{N}}^{\text{AE}}$ reaction

Space for Rough Work



Correct statement about (A), (B) and (C) is/are :

- (A) It is bromine water test of phenol in which white ppt is formed rapidly
 (B) It is the example of nucleophilic substitution reaction.
 (C) In water, phenol dissociate into H^+ and phenoxide ion and benzene ring in phenoxide ion is highly activated which repeatedly react with Br_2 in H_2O .
 (D) Overall it is electrophilic substitution reaction.
7. Lone pair of nitrogen is delocalised in :-
 (A) $(C_2H_5)_3N$ (B) $(SiH_3)_3N$ (C) H_3SiNCO (D) $H_2B - NH_2$
8. Identify reaction(s) which gives Homonuclear diatomic gas as one of the product.
 (A) Heating potassium chlorate (with MnO_2 catalyst)
 (B) Dissolving nitrogen dioxide in water
 (C) Heating NH_4NO_3 (below $300^\circ C$)
 (D) Oxidation of hydrogen chloride gas by oxygen in the presence of $CuCl_2$ catalyst

Space for Rough Work

9. Consider $[\text{Ni}(\text{L})_n]^{\pm x}$ (where Ni = nickel, L = ligand and n = number of ligand) and select **CORRECT** observation for complex.

(A) If L = NH_3 $n = 6 \Rightarrow \mu = \sqrt{8}$ B.M., outer orbital complex

(B) If L = Cl^- $n = 4 \Rightarrow \mu = \sqrt{8}$ B.M., high spin complex

(C) If L = $\begin{array}{c} \text{CH}_2\text{NH}_2 \\ | \\ \text{CH}_2\text{NH}_2 \end{array}$ $n = 3 \Rightarrow \mu = 0$, inner orbital complex

(D) If L = CN^- $n = 4 \Rightarrow \mu = 0$, low spin complex

10. For a chemical reaction : $\text{R} \rightarrow \text{P}$, the rate constant (k) is found to fit the following equation:

$\log (k \text{ (sec}^{-1}\text{)}) = 20 - \frac{6600}{T}$. Which of the following is/are correct for the reaction.

(A) Reaction follows first order kinetics

(B) Activation energy is approximately 55 kJ

(C) Half-life of reaction at 300 K is 69.30 sec

(D) The lowest possible limit of half life is 6.93×10^{-21} sec.

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

-									
●	●	●	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. A solution of palmitic acid (Molar mass = 256 gm) in benzene contains 5.12 gm of acid per litre. When this solution is dropped on a surface, benzene evaporates and palmitic acid forms a monolayer. If we wish to cover an area of 480 cm² with a monolayer what volume (in ml) of solution should be used ?
 {Area occupied by one molecules of palmitic acid = 0.2 nm²} {N_A = 6 × 10²³}
2. The specific conductivity of a saturated AgCl solution is 1.26 × 10⁻⁶ Scm⁻¹.

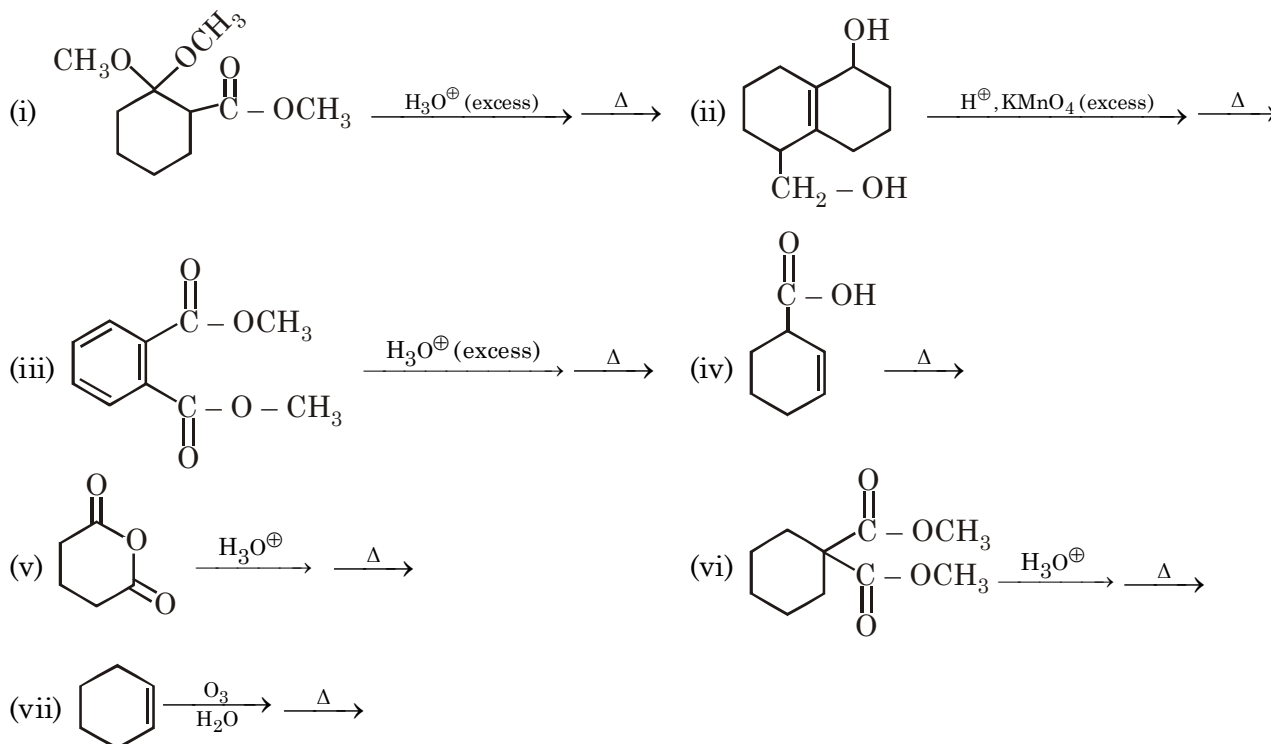
Given : $\Lambda_m^\infty(\text{Ag}^+) = 62 \text{ Scm}^2\text{mol}^{-1}$

$\Lambda_m^\infty(\text{Cl}^-) = 78 \text{ Scm}^2\text{mol}^{-1}$

If K_{sp} of AgCl is x × 10⁻¹¹, then value of 'x' is ?

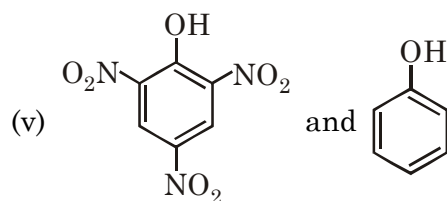
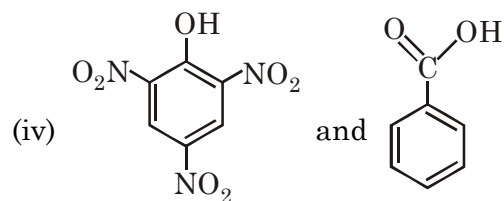
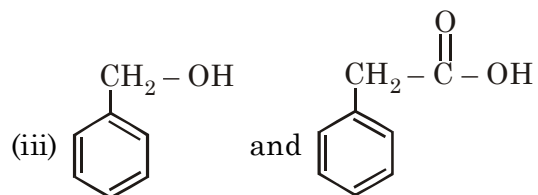
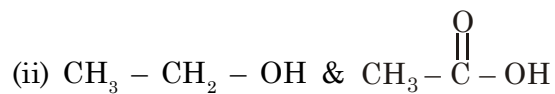
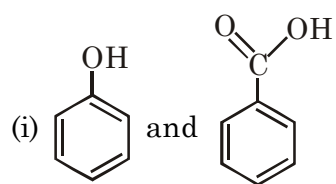
Space for Rough Work

3. What is the concentration in gm / litre of a solution of glucose which is isotonic with a deci-normal NaCl solution which is 85% dissociated ?
4. Calculate $|\Delta S_{\text{sys}}^0|$ (in J/K) for the following reaction at 300 K and 1 Bar.
 $\text{CO(g)} + \text{H}_2\text{O(g)} \rightarrow \text{CO}_2\text{(g)} + \text{H}_2\text{(g)}$
 Given for the reaction at 300 K :
 $\Delta H^0 = 2700 \text{ J}$
 $\Delta S_{\text{univ}}^0 = 45 \text{ JK}^{-1}$
5. How many reactions release CO_2 as one of the product on heating in reaction sequence?



Space for Rough Work

6. How many pair(s) of compounds is separated by both NaOH and NaHCO₃ :

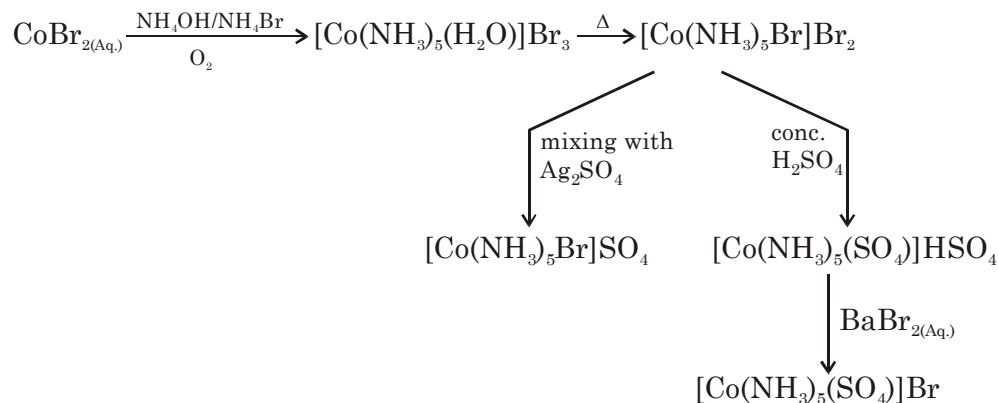


Space for Rough Work

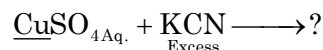
7. Consider different complexes form in given reaction sequence and find (X + Y).
 • X = Molecular weight of complex which forms in the step when oxidation number of cobalt is changed
 • Y = Molecular weight of complex whose ionisation isomer is also present in given reaction sequence

[Atomic weight. of Co = 59, Br = 80, S = 32, O = 16]

[Molecular weight of $\text{NH}_3 = 17$, $\text{H}_2\text{O} = 18$]



8. Find sum of the oxidation number of underlined metal in product of given three reactions.



Space for Rough Work

SECTION-IV : (Maximum Marks : 16)

- This section contains **TWO** questions.
- Each question contains two columns, **Column-I** and **Column-II**.
- **Column-I** has **four** entries (A), (B), (C) and (D)
- **Column-II** has **five** entries (P), (Q), (R), (S) and (T)
- Match the entries in **Column-I** with the entries in **column-II**.
- One or more entries in **Column-I** may match with one or more entries in **Column-II**.
- The ORS contains a 4×5 matrix whose layout will be similar to the one shown below :

(A)	☐ (P)	☐ (Q)	☐ (R)	☐ (S)	☐ (T)
(B)	☐ (P)	☐ (Q)	☐ (R)	☐ (S)	☐ (T)
(C)	☐ (P)	☐ (Q)	☐ (R)	☐ (S)	☐ (T)
(D)	☐ (P)	☐ (Q)	☐ (R)	☐ (S)	☐ (T)

- For each entry in **column-I**, darken the bubbles of all the matching entries. For example, if entry (A) in **Column-I** matches with entries (Q), (R) and (T), then darken these three bubbles in the ORS. Similarly, for entries (B), (C) and (D).

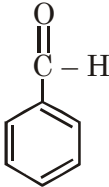
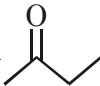
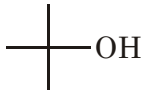
- For each question, marks will be awarded in one of the following categories :

For each entry in **Column-I**

Full Marks : +2 If only the bubble(s) corresponding to all the correct match(es) is (are) darkened

Zero Marks : 0 If none of the bubbles is darkened

Negative Marks : -1 In all other cases

1.	Column-I Pair of compounds	Column-II Differentiated by
(A)	$\text{H}-\overset{\text{O}}{\parallel}{\text{C}}-\text{H}$ and $\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{H}$	(P) $[\text{Ag}(\text{NH}_3)_2]\text{OH}$
(B)	 and 	(Q) 2,4-Dinitrophenyl hydrazine
(C)	$\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{H}$ and 	(R) NaOH , I_2 , Δ
(D)	 and $\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{H}$	(S) Fehling solution (T) Lucas reagent

Space for Rough Work

2. Match the following :-

Column - I

- (A) $\underline{\text{NO}}[\underline{\text{BF}}_4]$
 (B) $(\underline{\text{NH}}_4)_2 [\underline{\text{Ni}}\text{Cl}_4]$
 (C) CsO_2
 (D) $[\underline{\text{Co}}(\text{H}_2\text{O})_6][\underline{\text{Co}}\text{Cl}_4]$

Column - II

- (P) Spin magnetic moment $\mu \neq 0$, for cation
 (Q) Spin magnetic moment $\mu \neq 0$, for anion
 (R) sp^3 hybridised underline atom in cation
 (S) sp^3 hybridised underline atom in anion
 (T) π -bond(s) in cation

Space for Rough Work

PART-3 : MATHEMATICS

SECTION-I : (Maximum Marks: 40)

- This section contains **TEN** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Let $f : I \rightarrow I$ be a one-one function such that $f(0) = 1$, $f(f(n)) = f(f(n+2) + 2) = n$, then
 (A) $f(3) = 0$ (B) $f(2) = 0$ (C) $f(3) = -2$ (D) $f(2) = -1$
2. $\lim_{x \rightarrow \infty} \left(\frac{x+1}{x+2} \right)^x$ is equal to
 (A) $\frac{1}{e}$ (B) e (C) 1 (D) e^2
3. If $f(x) = [x]$ (sinkx)^p is continuous $\forall x \in D_f$, then ([.] represents the greatest integer function)
 (A) $k \in \{n\pi, n \in I\}, p \in R$ (B) $k \in \{2n\pi, n \in I\}, p \in R$
 (C) $k \in \{n\pi, n \in I\}, p \in R - \{0\}$ (D) $k \in \{n\pi, n \in I, n \neq 0\}, p \in R$

Space for Rough Work

4. Let $g(x) = e^{f(x)}$ and $f(x+1) = x + f(x) \forall x \in \mathbb{R}$. If $n \in \mathbb{N}$, then $\frac{g'\left(n + \frac{1}{2}\right)}{g\left(n + \frac{1}{2}\right)} - \frac{g'\left(\frac{1}{2}\right)}{g\left(\frac{1}{2}\right)}$ is always equal to
- (A) $2\left(1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n}\right)$ (B) $2\left(1 + \frac{1}{3} + \frac{1}{5} + \dots + \frac{1}{2n-1}\right)$
 (C) n (D) 1
5. Let $f(x)$ be a function such that its derivative $f'(x)$ is continuous in $[a, b]$ and differentiable in (a, b) . Consider a function $\phi(x) = f(b) - f(x) - (b-x)f'(x) - (b-x)^2A$. If Rolle's theorem is applicable to $\phi(x)$ on $[a, b]$ and there exist some number $c \in (a, b)$, such that $\phi'(c) = 0$ and $f(b) = f(a) + (b-a)f'(a) + \lambda(b-a)^2f''(c)$, then λ is
- (A) 0 (B) 1 (C) $\frac{1}{2}$ (D) $-\frac{1}{2}$
6. Let $f : \mathbb{R} \rightarrow \mathbb{R}$, $y = f(x)$, $f(0) = 0$, $f'(x) > 0$ and $f''(x) > 0$ three points $A(\alpha, f(\alpha))$, $B(\beta, f(\beta))$, $C(\gamma, f(\gamma))$ lie on $y = f(x)$ such that $0 < \alpha < \beta < \gamma$, then which of the following is/are true ?
- (A) $\alpha f(\beta) > \beta f(\alpha)$ (B) $\frac{f(\alpha) + f(\beta)}{2} > f\left(\frac{\alpha + \beta}{2}\right)$
 (C) $\gamma f(\gamma + \beta - \alpha) > (\gamma + \beta - \alpha)f(\gamma)$ (D) $\frac{f(\alpha) + f(\beta)}{2} < f\left(\frac{\alpha + \beta}{2}\right)$

Space for Rough Work

7. Consider the function $f(x) = \int_0^x (5\ln(1+t^2) - 10t \tan^{-1} t + 16 \sin t) dt$, then
- (A) $f(x)$ is increasing $\forall x \in (0,1)$ (B) $f(x)$ is decreasing $\forall x \in (0,1)$
- (C) $\int_0^x f(t) dt$ is positive $\forall x \in (0,1)$ (D) $\int_0^x f(t) dt$ is increasing for $x \in (0,1)$
8. The equations of the diagonals of the square formed by the pair of lines $xy + 4x - 3y - 12 = 0$ and $xy - 3x + 4y - 12 = 0$, is/are
- (A) $x - y = 0$ (B) $x + y + 1 = 0$ (C) $x + y = 0$ (D) $x - y + 1 = 0$
9. Consider the circles $\omega_1 : (x - a)^2 + (y - b)^2 = a^2 + b^2$ and $\omega_2 : (x - b)^2 + (y - a)^2 = a^2 + b^2$ (here a and b are distinct real numbers). Which of the following is/are true ?
- (A) The circles intersect at the origin and a point, which lie on line $y = 2x$.
- (B) The circles intersect at the origin and a point, which lie on line $y = 3x$.
- (C) The line $y = x$ is the radical axis of the pair of circles.
- (D) The circles are orthogonal for all distinct a and b .
10. P and Q are two points on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, whose eccentric angles are differ by 90° , then
- (A) Locus of point of intersection of tangents at P and Q is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 2$
- (B) Locus of mid point of P and Q is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{1}{2}$
- (C) Locus of point of intersection of tangents at P and Q is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{1}{2}$
- (D) Locus of midpoint of P and Q is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 2$

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	○	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	●	●	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

-									
●	●	●	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	⑤	●	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. $x^3 + 5x^2 + px + q = 0$ and $x^3 + 7x^2 + px + r = 0$, have two roots in common. If their third roots are γ_1 and γ_2 respectively, then $|\gamma_1 + \gamma_2|$ is equal to
2. Let $|z - (1 + i)| = 2\sqrt{2}$ then maximum value of $[|z|]$
(Where $[.]$ is greatest integer function)
3. Let d is the distance of a plane P from the point (2,1,0). Plane P passes through (1,2,1) and contains a line $x + 2y + z = 0 = 2x - y - z - 1$ then $27d^2$ is

Space for Rough Work

4. The number of pairs of diagonals of a regular polygon of 10 sides that are parallel, are
5. If P_n denotes the product of all the coefficient in the expansion of $(1 + x)^n$ and $\frac{P_n}{P_{n+1}} = 10^9$, then n is equal to ($n \in \mathbb{N}$)
6. If 'A' is an idempotent matrix satisfying $(I - 0.4A)^{-1} = (I - \alpha A)$ (where I is the unit matrix of same order as that of 'A', A is not a null matrix), then $\left| \frac{1}{\alpha} \right|$ is
-

Space for Rough Work

7. If a and b are randomly chosen from the set $\{1, 2, 3, 4, 5, 6, 7, 8, 9\}$, then the probability that the expression ' $ax^4 + bx^3 + (a + 1)x^2 + bx + 1$ ' has positive values for $\forall x \in \mathbb{R}$ is $\frac{p}{q}$ (p & q are coprime numbers), then $q - p$ is
8. If $\vec{a}, \vec{b}$ are vectors perpendicular to each other and $|\vec{a}| = 2, |\vec{b}| = 3, \vec{c} \times \vec{a} = \vec{b}$, then the least value of $2|\vec{c} - \vec{a}|$ is

Space for Rough Work

SECTION-IV : (Maximum Marks : 16)

- This section contains **TWO** questions.
- Each question contains two columns, **Column-I** and **Column-II**.
- **Column-I** has **four** entries (A), (B), (C) and (D)
- **Column-II** has **five** entries (P), (Q), (R), (S) and (T)
- Match the entries in **Column-I** with the entries in **column-II**.
- One or more entries in **Column-I** may match with one or more entries in **Column-II**.
- The ORS contains a 4×5 matrix whose layout will be similar to the one shown below :

(A)	☐ (P)	☐ (Q)	☐ (R)	☐ (S)	☐ (T)
(B)	☐ (P)	☐ (Q)	☐ (R)	☐ (S)	☐ (T)
(C)	☐ (P)	☐ (Q)	☐ (R)	☐ (S)	☐ (T)
(D)	☐ (P)	☐ (Q)	☐ (R)	☐ (S)	☐ (T)

- For each entry in **column-I**, darken the bubbles of all the matching entries. For example, if entry (A) in **Column-I** matches with entries (Q), (R) and (T), then darken these three bubbles in the ORS. Similarly, for entries (B), (C) and (D).
- For each question, marks will be awarded in one of the following categories :

For each entry in **Column-I**

Full Marks : +2 If only the bubble(s) corresponding to all the correct match(es) is (are) darkened

Zero Marks : 0 If none of the bubbles is darkened

Negative Marks : -1 In all other cases

1. In a $\triangle ABC$, a , b and A are given $b > a$ and c_1 , c_2 are two possible values of the third side 'c'. If Δ_1 and Δ_2 are areas of two triangles with sides a , b , c_1 and a , b , c_2 , then

Column-I	Column-II
(A) Arithmetic mean of Δ_1 and Δ_2 is	(P) $\left(\frac{b^2 - a^2}{2}\right) \tan A$
(B) Geometric mean of Δ_1 and Δ_2 is	(Q) $\frac{1}{4} b^2 \sin 2A$
(C) Harmonic mean of Δ_1 and Δ_2 is	(R) $\frac{1}{2} b \sin A \left(\sqrt{b^2 - a^2}\right)$
(D) $ \Delta_1 - \Delta_2 $ is equal to	(S) $b \sin A$
	(T) $b \sin A \sqrt{a^2 - b^2 \sin^2 A}$

Space for Rough Work

2. The system of equation

$$x + y + z = 3$$

$$x + 2y + 3z = 6$$

$$x + 3y + \lambda z = m,$$

Column-I

- (A) has unique solution
- (B) has no solution
- (C) has infinite number of solution
- (D) is consistent

Column-II

- (P) If $\lambda = 5$, $m = 9$
- (Q) If $\lambda = 5$, $m \neq 9$
- (R) If $\lambda \neq 5$, $m \neq 9$
- (S) If $\lambda \neq 5$, $m \in \mathbb{R}$
- (T) If $\lambda \neq 5$, $m = 9$

Space for Rough Work